

## Matrices and Systems of Linear Equations (2015–2021)

Category: Linear Algebra and Algebra • Official Mock Test Series

TOTAL QUESTIONS

34

TOTAL MARKS

60 M

TEST DURATION

102 Min

PAPER PATTERN

MCQ • MSQ • NAT  
(2015–20)

| Section / Type         | Questions | Marking Scheme | Negative Marking  | Format / Details       |
|------------------------|-----------|----------------|-------------------|------------------------|
| Multiple Choice (MCQ)  | 16        | +1 / +2 Marks  | −1/3 / −2/3 Marks | 4 Options (1 Correct)  |
| Multiple Select (MSQ)  | 6         | +2 Marks       | Nil (0)           | 4 Options (1+ Correct) |
| Numerical Answer (NAT) | 12        | +1 / +2 Marks  | Nil (0)           | Real Decimal Value     |

## IMPORTANT INSTRUCTIONS FOR CANDIDATES

- Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
- Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
- Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
- Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
- Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
- Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at [vaibhavgeometer.github.io](https://vaibhavgeometer.github.io).

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2021 • Q59 • ID: JAM\_2021\_Q59)

NAT

+2 M

Neg: Nil (0)

Q. 59 Let  $A = \begin{pmatrix} 2 & -1 & 3 \\ 2 & -1 & 3 \\ 3 & 2 & -1 \end{pmatrix}$ . Then the largest eigenvalue of  $A$  is \_\_\_\_.

Question 2 (JAM 2021 • Q56 • ID: JAM\_2021\_Q56)

NAT

+2 M

Neg: Nil (0)

JAM 2021

MATHEMATICS - MA

Q. 56 The determinant of the matrix

$$\begin{pmatrix} 2021 & 2020 & 2020 & 2020 \\ 2021 & 2021 & 2020 & 2020 \\ 2021 & 2021 & 2021 & 2020 \\ 2021 & 2021 & 2021 & 2021 \end{pmatrix}$$

is \_\_\_\_.

Question 3 (JAM 2021 • Q40 • ID: JAM\_2021\_Q40)

MSQ

+2 M

Neg: Nil (0)

Q. 40 Let  $m > 1$  and  $n > 1$  be integers. Let  $A$  be an  $m \times n$  matrix such that for some  $m \times 1$  matrix  $b_1$ , the equation  $Ax = b_1$  has infinitely many solutions. Let  $b_2$  denote an  $m \times 1$  matrix different from  $b_1$ . Then  $Ax = b_2$  has

- (A) infinitely many solutions for some  $b_2$ .      (B) a unique solution for some  $b_2$ .  
 (C) no solution for some  $b_2$ .      (D) finitely many solutions for some  $b_2$ .



MA

13 / 17

Question 4 (JAM 2021 • Q22 • ID: JAM\_2021\_Q22)

MCQ

+2 M

Neg: -0.67

Q. 22 Let  $A$  be an  $n \times n$  invertible matrix and  $C$  be an  $n \times n$  nilpotent matrix. If  $X = \begin{pmatrix} X_{11} & X_{12} \\ X_{21} & X_{22} \end{pmatrix}$  is a  $2n \times 2n$  matrix (each  $X_{ij}$  being  $n \times n$ ) that commutes with the  $2n \times 2n$  matrix  $B = \begin{pmatrix} A & 0 \\ 0 & C \end{pmatrix}$ , then

- (A)  $X_{11}$  and  $X_{22}$  are necessarily zero matrices.
- (B)  $X_{12}$  and  $X_{21}$  are necessarily zero matrices.
- (C)  $X_{11}$  and  $X_{21}$  are necessarily zero matrices.
- (D)  $X_{12}$  and  $X_{22}$  are necessarily zero matrices.

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Question 5 (JAM 2021 • Q10 • ID: JAM\_2021\_Q10)

MCQ

+1 M

Neg: -0.33

JAM 2021

MATHEMATICS - MA

Q. 10 Let  $n > 1$  be an integer. Consider the following two statements for an arbitrary  $n \times n$  matrix  $A$  with complex entries.

- I. If  $A^k = I_n$  for some integer  $k \geq 1$ , then all the eigenvalues of  $A$  are  $k^{\text{th}}$  roots of unity.  
II. If, for some integer  $k \geq 1$ , all the eigenvalues of  $A$  are  $k^{\text{th}}$  roots of unity, then  $A^k = I_n$ .

Then

- (A) both I and II are TRUE. (B) I is TRUE but II is FALSE.  
(C) I is FALSE but II is TRUE. (D) neither I nor II is TRUE.



MA

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Question 6 (JAM 2020 • Q58 • ID: JAM\_2020\_Q58)

NAT

+2 M

Neg: Nil (0)

Q. 58 Let  $M = \begin{bmatrix} 9 & 2 & 7 & 1 \\ 0 & 7 & 2 & 1 \\ 0 & 0 & 11 & 6 \\ 0 & 0 & -5 & 0 \end{bmatrix}$ . Then, the value of  $\det((8I - M)^3)$  is \_\_\_\_\_.

Question 7 (JAM 2020 • Q36 • ID: JAM\_2020\_Q36)

MSQ

+2 M

Neg: Nil (0)

Q. 36 Consider the following system of linear equations

$$x + y + 5z = 3, \quad x + 2y + mz = 5 \quad \text{and} \quad x + 2y + 4z = k.$$

The system is consistent if

(A)  $m \neq 4$ (B)  $k \neq 5$ (C)  $m = 4$ (D)  $k = 5$ 

MA

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Question 8 (JAM 2020 • Q20 • ID: JAM\_2020\_Q20)

MCQ

+2 M

Neg: -0.67

Q. 20 Let  $M$  be an  $n \times n$  ( $n \geq 2$ ) non-zero real matrix with  $M^2 = 0$  and let  $\alpha \in \mathbb{R} \setminus \{0\}$ . Then

(A)  $\alpha$  is the only eigenvalue of  $(M + \alpha I)$  and  $(M - \alpha I)$ (B)  $\alpha$  is the only eigenvalue of  $(M + \alpha I)$  and  $(\alpha I - M)$ (C)  $-\alpha$  is the only eigenvalue of  $(M + \alpha I)$  and  $(M - \alpha I)$ (D)  $-\alpha$  is the only eigenvalue of  $(M + \alpha I)$  and  $(\alpha I - M)$ 

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Question 9 (JAM 2020 • Q16 • ID: JAM\_2020\_Q16)

MCQ

+2 M

Neg: -0.67

JAM 2020

MATHEMATICS - MA

Q. 16 Let  $S$  be the surface of the portion of the sphere with centre at the origin and radius 4, above the  $xy$ -plane. Let  $\vec{F} = y\hat{i} - x\hat{j} + yx^3\hat{k}$ . If  $\hat{n}$  is the unit outward normal to  $S$ , then

$$\iint_S (\nabla \times \vec{F}) \cdot \hat{n} \, dS$$

equals

(A)  $-32\pi$ (B)  $-16\pi$ (C)  $16\pi$ (D)  $32\pi$ 

Question 10 (JAM 2020 • Q5 • ID: JAM\_2020\_Q5)

MCQ

+1 M

Neg: -0.33

JAM 2020

MATHEMATICS - MA

Q. 5 If the equation of the tangent plane to the surface  $z = 16 - x^2 - y^2$  at the point  $P(1, 3, 6)$  is  $ax + by + cz + d = 0$ , then the value of  $|d|$  is

(A) 16

(B) 26

(C) 36

(D) 46

Question 11 (JAM 2019 • Q58 • ID: JAM\_2019\_Q58)

NAT

+2 M

Neg: Nil (0)

Q.58 Let  $M$  be a  $3 \times 3$  matrix with real entries such that  $M^2 = M + 2I$ , where  $I$  denotes the  $3 \times 3$  identity matrix. If  $\alpha, \beta$  and  $\gamma$  are eigenvalues of  $M$  such that  $\alpha\beta\gamma = -4$ , then  $\alpha + \beta + \gamma$  is equal to \_\_\_\_\_

Question 12 (JAM 2019 • Q57 • ID: JAM\_2019\_Q57)

NAT

+2 M

Neg: Nil (0)

Q.57 Let  $M$  and  $N$  be any two  $4 \times 4$  matrices with integer entries satisfying

$$MN = 2 \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Then the maximum value of  $\det(M) + \det(N)$  is \_\_\_\_\_

Question 13 (JAM 2019 • Q56 • ID: JAM\_2019\_Q56)

NAT

+2 M

Neg: Nil (0)

Q.56 If  $\begin{pmatrix} 2 \\ y \\ z \end{pmatrix}$ ,  $y, z \in \mathbb{R}$ , is an eigenvector corresponding to a real eigenvalue of the matrix  $\begin{pmatrix} 0 & 0 & 2 \\ 1 & 0 & -4 \\ 0 & 1 & 3 \end{pmatrix}$  then  $z - y$  is equal to \_\_\_\_\_

Question 14 (JAM 2019 • Q43 • ID: JAM\_2019\_Q43)

NAT

+1 M

Neg: Nil (0)

Q.43 Consider the following system of three linear equations in four unknowns  $x_1, x_2, x_3$  and  $x_4$ 

$$\begin{aligned} x_1 + x_2 + x_3 + x_4 &= 4, \\ x_1 + 2x_2 + 3x_3 + 4x_4 &= 5, \\ x_1 + 3x_2 + 5x_3 + kx_4 &= 5. \end{aligned}$$

If the system has no solutions, then  $k =$  \_\_\_\_\_

Question 15 (JAM 2019 • Q27 • ID: JAM\_2019\_Q27)

MCQ

+2 M

Neg: -0.67

Q.27 The set of eigenvalues of which one of the following matrices is NOT equal to the set of eigenvalues of  $\begin{pmatrix} 1 & 2 \\ 4 & 3 \end{pmatrix}$ ?

(A)  $\begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}$

(B)  $\begin{pmatrix} 3 & 2 \\ 4 & 1 \end{pmatrix}$

(C)  $\begin{pmatrix} 3 & 4 \\ 2 & 1 \end{pmatrix}$

(D)  $\begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix}$

Question 16 (JAM 2019 • Q16 • ID: JAM\_2019\_Q16)

MCQ

+2 M

Neg: -0.67

Q.16 Let  $f(x) = (\ln x)^2$ ,  $x > 0$ . Then

- (A)  $\lim_{x \rightarrow \infty} \frac{f(x)}{x}$  does not exist  
 (B)  $\lim_{x \rightarrow \infty} f'(x) = 2$   
 (C)  $\lim_{x \rightarrow \infty} (f(x+1) - f(x)) = 0$   
 (D)  $\lim_{x \rightarrow \infty} (f(x+1) - f(x))$  does not exist

Question 17 (JAM 2019 • Q5 • ID: JAM\_2019\_Q5)

MCQ

+1 M

Neg: -0.33

Q.5 Let  $S$  be the set of all limit points of the set  $\left\{\frac{n}{\sqrt{2}} + \frac{\sqrt{2}}{n} : n \in \mathbb{N}\right\}$ . Let  $\mathbb{Q}_+$  be the set of all positive rational numbers. Then

- (A)  $\mathbb{Q}_+ \subseteq S$  (B)  $S \subseteq \mathbb{Q}_+$   
 (C)  $S \cap (\mathbb{R} \setminus \mathbb{Q}_+) \neq \emptyset$  (D)  $S \cap \mathbb{Q}_+ \neq \emptyset$

MA

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Question 18 (JAM 2018 • Q37 • ID: JAM\_2018\_Q37)

MSQ

+2 M

Neg: Nil (0)

Q.37 Let  $m, n \in \mathbb{N}$ ,  $m < n$ ,  $P \in M_{n \times m}(\mathbb{R})$ ,  $Q \in M_{m \times n}(\mathbb{R})$ . Then which of the following is (are) NOT possible?

- (A)  $\text{rank}(PQ) = n$   
 (B)  $\text{rank}(QP) = m$   
 (C)  $\text{rank}(PQ) = m$   
 (D)  $\text{rank}(QP) = \left\lceil \frac{m+n}{2} \right\rceil$ , the smallest integer larger than or equal to  $\frac{m+n}{2}$



Question 19 (JAM 2018 • Q36 • ID: JAM\_2018\_Q36)

MSQ

+2 M

Neg: Nil (0)

Q.36 Suppose  $\alpha, \beta, \gamma \in \mathbb{R}$ . Consider the following system of linear equations.

$x + y + z = \alpha$ ,  $x + \beta y + z = \gamma$ ,  $x + y + \alpha z = \beta$ . If this system has at least one solution, then which of the following statements is (are) TRUE?

- (A) If  $\alpha = 1$  then  $\gamma = 1$  (B) If  $\beta = 1$  then  $\gamma = \alpha$   
 (C) If  $\beta \neq 1$  then  $\alpha = 1$  (D) If  $\gamma = 1$  then  $\alpha = 1$

Question 20 (JAM 2018 • Q27 • ID: JAM\_2018\_Q27)

MCQ

+2 M

Neg: -0.67

Q.27 Let  $I$  denote the  $4 \times 4$  identity matrix. If the roots of the characteristic polynomial of a  $4 \times 4$  matrix

$M$  are  $\pm \sqrt{\frac{1 \pm \sqrt{5}}{2}}$ , then  $M^8 =$

- (A)  $I + M^2$  (B)  $2I + M^2$  (C)  $2I + 3M^2$  (D)  $3I + 2M^2$

Question 21 (JAM 2018 • Q17 • ID: JAM\_2018\_Q17)

MCQ

+2 M

Neg: -0.67

Q.17 Let  $f(x, y) = \begin{cases} \frac{xy}{(x^2+y^2)^\alpha}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$

Then which one of the following is TRUE for  $f$  at the point  $(0, 0)$ ?

- (A) For  $\alpha = 1$ ,  $f$  is continuous but not differentiable  
 (B) For  $\alpha = \frac{1}{2}$ ,  $f$  is continuous and differentiable  
 (C) For  $\alpha = \frac{1}{4}$ ,  $f$  is continuous and differentiable  
 (D) For  $\alpha = \frac{3}{4}$ ,  $f$  is neither continuous nor differentiable

Question 22 (JAM 2018 • Q7 • ID: JAM\_2018\_Q7)

MCQ

+1 M

Neg: -0.33

Q.7 Let  $f: \mathbb{R}^3 \rightarrow \mathbb{R}$  be a scalar field,  $\vec{v}: \mathbb{R}^3 \rightarrow \mathbb{R}^3$  be a vector field and let  $\vec{a} \in \mathbb{R}^3$  be a constant vector.

If  $\vec{r}$  represents the position vector  $x\hat{i} + y\hat{j} + z\hat{k}$ , then which one of the following is FALSE?

(A)  $\text{curl}(f \vec{v}) = \text{grad}(f) \times \vec{v} + f \text{curl}(\vec{v})$

(B)  $\text{div}(\text{grad}(f)) = \left( \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) f$

(C)  $\text{curl}(\vec{a} \times \vec{r}) = 2 |\vec{a}| \vec{r}$

(D)  $\text{div}\left(\frac{\vec{r}}{|\vec{r}|^3}\right) = 0$ , for  $\vec{r} \neq \vec{0}$

Question 23 (JAM 2017 • Q58 • ID: JAM\_2017\_Q58)

NAT

+2 M

Neg: Nil (0)

Q.58 Let  $\alpha, \beta, \gamma, \delta$  be the eigenvalues of the matrix

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix}$$

Then  $\alpha^2 + \beta^2 + \gamma^2 + \delta^2 = \underline{\hspace{2cm}}$

Question 24 (JAM 2017 • Q48 • ID: JAM\_2017\_Q48)

NAT

+1 M

Neg: Nil (0)

Q.48 Let  $P$  be a  $7 \times 7$  matrix of rank 4 with real entries. Let  $\mathbf{a} \in \mathbb{R}^7$  be a column vector. Then the rank of  $P + \mathbf{a}\mathbf{a}^T$  is at least  $\underline{\hspace{2cm}}$

Question 25 (JAM 2017 • Q38 • ID: JAM\_2017\_Q38)

MSQ

+2 M

Neg: Nil (0)

Q.38 Let  $M$  be an  $n \times n$  matrix with real entries such that  $M^3 = I$ . Suppose that  $Mv \neq v$  for any non-zero vector  $v$ . Then which of the following statements is / are TRUE?

- (A)  $M$  has real eigenvalues
- (B)  $M + M^{-1}$  has real eigenvalues
- (C)  $n$  is divisible by 2
- (D)  $n$  is divisible by 3

Question 26 (JAM 2017 • Q34 • ID: JAM\_2017\_Q34)

MSQ

+2 M

Neg: Nil (0)

Q.34 If  $X$  and  $Y$  are  $n \times n$  matrices with real entries, then which of the following is/are TRUE?

- (A) If  $P^{-1}XP$  is diagonal for some real invertible matrix  $P$ , then there exists a basis for  $\mathbb{R}^n$  consisting of eigenvectors of  $X$
- (B) If  $X$  is diagonal with distinct diagonal entries and  $XY = YX$ , then  $Y$  is also diagonal
- (C) If  $X^2$  is diagonal, then  $X$  is diagonal
- (D) If  $X$  is diagonal and  $XY = YX$  for all  $Y$ , then  $X = \lambda I$  for some  $\lambda \in \mathbb{R}$

Question 27 (JAM 2017 • Q12 • ID: JAM\_2017\_Q12)

MCQ

+2 M

Neg: -0.67

Q.12 Let  $\mathcal{M}$  be the set of all invertible  $5 \times 5$  matrices with entries 0 and 1. For each  $M \in \mathcal{M}$ , let  $n_1(M)$  and  $n_0(M)$  denote the number of 1's and 0's in  $M$ , respectively. Then

$$\min_{M \in \mathcal{M}} |n_1(M) - n_0(M)| =$$

- (A) 1
- (B) 3
- (C) 5
- (D) 15

Question 28 (JAM 2016 • Q53 • ID: JAM\_2016\_Q53)

NAT

+2 M

Neg: Nil (0)

Q.53 Let

$$M = \begin{bmatrix} \alpha & 1 & 1 \\ 1 & \beta & 1 \\ 1 & 1 & \gamma \end{bmatrix}, \quad \alpha\beta\gamma = 1, \quad \alpha, \beta, \gamma \in \mathbb{R} \quad \text{and} \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3.$$

Then  $M\mathbf{x} = \mathbf{0}$  has infinitely many solutions if  $\text{trace}(M)$  is \_\_\_\_\_

Question 29 (JAM 2016 • Q43 • ID: JAM\_2016\_Q43)

NAT

+1 M

Neg: Nil (0)

Q.43 Let  $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3$  be a non-zero vector and  $A = \frac{\mathbf{x}\mathbf{x}^T}{\mathbf{x}^T\mathbf{x}}$ . Then the dimension of the vector space  $\left\{ \mathbf{y} \in \mathbb{R}^3 \mid A\mathbf{y} = \mathbf{0} \right\}$  over  $\mathbb{R}$  is \_\_\_\_\_

Question 30 (JAM 2016 • Q26 • ID: JAM\_2016\_Q26)

MCQ

+2 M

Neg: -0.67

Q.26 The matrix  $M = \begin{bmatrix} \cos \alpha & \sin \alpha \\ i \sin \alpha & i \cos \alpha \end{bmatrix}$  is a unitary matrix when  $\alpha$  is

(A)  $(2n+1)\frac{\pi}{2}, n \in \mathbb{Z}$

(B)  $(3n+1)\frac{\pi}{3}, n \in \mathbb{Z}$

(C)  $(4n+1)\frac{\pi}{4}, n \in \mathbb{Z}$

(D)  $(5n+1)\frac{\pi}{5}, n \in \mathbb{Z}$

Question 31 (JAM 2016 • Q16 • ID: JAM\_2016\_Q16)

MCQ

+2 M

Neg: -0.67

Q.16 The surface area of the portion of the plane  $y + 2z = 2$  within the cylinder  $x^2 + y^2 = 3$  is

(A)  $\frac{3\sqrt{5}}{2}\pi$

(B)  $\frac{5\sqrt{5}}{2}\pi$

(C)  $\frac{7\sqrt{5}}{2}\pi$

(D)  $\frac{9\sqrt{5}}{2}\pi$

Question 32 (JAM 2015 • Q60 • ID: JAM\_2015\_Q60)

NAT

+2 M

Neg: Nil (0)

Q.20 Let  $P$  and  $Q$  be two real matrices of size  $4 \times 6$  and  $5 \times 4$ , respectively. If  $\text{rank}(Q) = 4$  and  $\text{rank}(QP) = 2$ , then  $\text{rank}(P)$  is equal to \_\_\_\_\_

**END OF THE QUESTION PAPER**

MA

16/16

Question 33 (JAM 2015 • Q16 • ID: JAM\_2015\_Q16)

MCQ

+2 M

Neg: -0.67

Q.16

Let  $A = \begin{bmatrix} 0 & 1-i \\ -1-i & i \end{bmatrix}$  and  $B = A^T \bar{A}$ . Then

- (A) an eigenvalue of  $B$  is purely imaginary
- (B) an eigenvalue of  $A$  is zero
- (C) all eigenvalues of  $B$  are real
- (D)  $A$  has a non-zero real eigenvalue

Question 34 (JAM 2015 • Q5 • ID: JAM\_2015\_Q5)

MCQ

+1 M

Neg: -0.33

Q.5 Let  $\{x_n\}$  be a convergent sequence of real numbers. If  $x_1 > \pi + \sqrt{2}$  and  $x_{n+1} = \pi + \sqrt{x_n - \pi}$  for  $n \geq 1$ , then which one of the following is the limit of this sequence?

- (A)  $\pi + 1$                       (B)  $\pi + \sqrt{2}$                       (C)  $\pi$                       (D)  $\pi + \sqrt{\pi}$

**OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC**

Matrices and Systems of Linear Equations (2015–2021) • All Official Answers &amp; Acceptance Ranges

| Q. No. | Type | Marks | Official Key |
|--------|------|-------|--------------|
| 1      | NAT  | +2    | 4            |
| 2      | NAT  | +2    | 2021         |
| 3      | MSQ  | +2    | MTA          |
| 4      | MCQ  | +2    | B            |
| 5      | MCQ  | +1    | B            |
| 6      | NAT  | +2    | −216 to −216 |
| 7      | MSQ  | +2    | A;D          |
| 8      | MCQ  | +2    | B            |
| 9      | MCQ  | +2    | A            |
| 10     | MCQ  | +1    | Marks to All |
| 11     | NAT  | +2    | 3 to 3       |
| 12     | NAT  | +2    | 17 to 17     |

| Q. No. | Type | Marks | Official Key |
|--------|------|-------|--------------|
| 13     | NAT  | +2    | 3 to 3       |
| 14     | NAT  | +1    | 7 to 7       |
| 15     | MCQ  | +2    | D            |
| 16     | MCQ  | +2    | C            |
| 17     | MCQ  | +1    | B            |
| 18     | MSQ  | +2    | A;D          |
| 19     | MSQ  | +2    | A;B          |
| 20     | MCQ  | +2    | C            |
| 21     | MCQ  | +2    | C            |
| 22     | MCQ  | +1    | C            |
| 23     | NAT  | +2    | 5.9 to 6.1   |
| 24     | NAT  | +1    | 2.9 to 3.1   |

| Q. No. | Type | Marks | Official Key |
|--------|------|-------|--------------|
| 25     | MSQ  | +2    | B, C         |
| 26     | MSQ  | +2    | A, B, D      |
| 27     | MCQ  | +2    | A            |
| 28     | NAT  | +2    | 3.0 to 3.0   |
| 29     | NAT  | +1    | 2.0 to 2.0   |
| 30     | MCQ  | +2    | A            |
| 31     | MCQ  | +2    | A            |
| 32     | NAT  | +2    | 2            |
| 33     | MCQ  | +2    | C            |
| 34     | MCQ  | +1    | A            |

**SCORING METHODOLOGY & EVALUATION GUIDELINES**

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt ( $-1/3$  for 1-mark questions,  $-2/3$  for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g.,  $[a, b]$ ) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.